

Assignment 6 - Wi-Fi

10.3.2024

Sent

100%

Forsøk 2



Se gjennom tilbakemelding

9.4.2024

Forsøk 2 Poengsum:

100%



Vis tilbakemelding

Anonym vurdering: **nei**

Ubegrenset antall forsøk tillatt

16.12.2023 til 29.4.2024

▼ Detaljer

Preparation

This assignment assumes you have completed all steps of the previous assignments. The resources and general instructions from previous assignments are still useful here, this assignment will only list new relevant resources. However, there is one major change; from this assignment and on, we will leave bare metal projects and move to full Mbed OS projects.

1. Create a new program for target DISCO-L475VG-IOT01A (B-L475E-IOT01A) under your git folder (`<course folder>/assignments/`) called `assignment_6`. If you create this project based on an example pick `mbed-os-example-blinky`. If you just copy a previous assignment, make sure the `mbed_app.json` file (if it exists) does not contain the line:
`"requires": ["bare-metal"],`
2. In this assignment we will use WiFi, but not other libraries for network connectivity (BLE, LoRaWAN, Cellular, NFC), and more, so you might consider shrinking compile time by downloading this [.mbedignore \(https://uia.instructure.com/courses/14984/files/2350766?wrap=1\)](https://uia.instructure.com/courses/14984/files/2350766?wrap=1)  (https://uia.instructure.com/courses/14984/files/2350766/download?download_frd=1) file, unzip and add it next to your `main.cpp` file in the project.
3. Add Wi-Fi network configuration for your local network to `mbed_app.json` as described in the lecture.

Resources

- See the lecture for examples
- [Wi-Fi error codes](https://os.mbed.com/docs/mbed-os/v6.16/mbed-os-api-doxy/nsapi__types_8h_source.html)  (https://os.mbed.com/docs/mbed-os/v6.16/mbed-os-api-doxy/nsapi__types_8h_source.html)
- Niels Lohmann JSON for modern C++ ([json.hpp file \(https://uia.instructure.com/courses/14984/files/2350759?wrap=1\)](https://uia.instructure.com/courses/14984/files/2350759?wrap=1)  (https://uia.instructure.com/courses/14984/files/2350759/download?download_frd=1) to be added next to your `main.cpp` file)
- [Niels Lohmann JSON for modern C++ examples](https://github.com/nlohmann/)  (<https://github.com/nlohmann/>)

Requirements

In this assignment you will create a program that connects to a server over Wi-Fi to download some data.

1. 10% Connect to your network. If you are at the lab you can use the **student-FoU** network.

The password is **stud2018**

Print a log message saying if the connection was successful or not.

2. 20% Connect to the following server: www.mocky.io ➞ <http://www.mocky.io>

Print a log message saying if the connection was successful or not.

3. 20% Send a HTTP GET request for the following resource: /

`v3/8dc9d952-8937-4eec-978b-40bc0a5daba1`

Note: Use www.mocky.io as Host in the request. Host is the FQDN only, protocol (HTTP) and resource `/v3/..` is not part of the host.

Update 2024-04-03: It seems that [mocky.io](http://www.mocky.io) is not currently handling HTTP GET requests to custom made endpoints, so try server ikt104.andersenitc.no ➞ [http://](http://ikt104.andersenitc.no)

ikt104.andersenitc.no and the resource `/index.json`

4. 20% Call `socket.recv()` function with a (chunk) size of 100 bytes. Store the received data in a character array and print the response from the server while reading it. Each chunk separated by three dots (...).
5. 10% Then after the complete response is received, print this array.
6. 20% The response is actually JSON formatted. Use a JSON library to parse the response, then: Print out first name, last name and age from the parsed data.

Note: This requirement is more difficult than the rest and should be attempted last. Even without this step you will get 80% for the rest.

Delivery

Deliver the following as text in Canvas:

- Link to Bitbucket where the code has been delivered.
Directly to the file if there is only 1, or the directory if more than 1.
- Link to YouTube (or other video service) with video showing each of the requirements.
Make sure that each requirement you have completed is clearly shown.
You don't need to show the actual code, just which of the requirements are met.